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Adaptive temporal graph neural networks for detecting coordinated multi-stage cyber-attacks in enterprise systems

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Abstract

This paper presents the problem of detecting complex, coordinated cyber-attacks at multiple intervals across enterprise networks. Traditional intrusion detection systems lack the ability to identify the 'when' and 'where' of attacks, resulting in an incomplete understanding of an attack's propagation. Most current methodologies focus on the analysis of single, disconnected, and isolated events. This fundamentally limits their ability and effectiveness to understand or conceptualise complex attacks that develop or evolve over time. This paper presents a framework composed of GNNs that captures and analyses the activity of enterprises as dynamic interaction graphs, applying a temporal attention mechanism to focus and prioritise the analysis of high-risk, risky behavioural patterns. The framework is tested against the LAANL Logs and DARPA Transparent Computing cybersecurity datasets. The proposed adaptive temporal GNN achieves a 5% improvement over the strongest baseline (GAT) in detection accuracy, demonstrating its effectiveness in capturing multi-stage coordinated cyber-attacks. Furthermore, the framework provided a consistent and significant improvement in test environments that were imposed with noise. Lastly, the framework was also able to improve explainability through the analysis of the path of an attack's propagation. This suggests that the results of the framework will validate the radical combination of the structural and the temporal elements of a complex cyber attack MODEL as a possible answer to the challenge of providing a simple, dynamic, and effective scalable solution to the detection of sophisticated cyber disasters in realistic and complex environments.

Keywords Graph Neural Networks, Intrusion detection, Multi-stage cyber-attacks, Temporal attention, Anomaly detection, Dynamic graph learning, Process innovation

1 Introduction

As cloud computing, the Internet of Things (IoT), and dispersed information systems are adopted increasingly, enterprise digital infrastructures rapidly transform [14]. Indeed, changes in scalability and operational efficiency are positive, and the interconnected and complex surfaces for attacks are negative. Unfolding of cyber-attacks has changed from elaborate single incidents and attacks into multiple, coordinated, complex attacks, and

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multi-pronged attacks involving reconnaissance, lateral movement, privilege escalation, and data extraction, from and across dispersed attack environments [13]. Such attacks are stealthy and dynamic and, in fact, time-dependent, and therefore, the challenge in detection is significantly higher than in ordinary single-stage attacks.

Most traditional, machine learning-based, and signature-based intrusion detection systems (IDS) lack the capability of detecting increasingly sophisticated attacks [1]. Most of these systems have been designed for detecting attacks in isolation; therefore, they fail to understand the coordinated attack context and the time-dependent progression of attacks. These systems, therefore, ultimately lead to diminishing returns in terms of attack detection and have a propensity of being falsely-diagnosed. Sophisticated attacks often go undetected, and while advanced machine learning systems based on deep learning have been known for sophisticated representation learning, they have not been used for sophisticated attacks to date, especially in enterprise systems, where modern attacks have created increasingly complex networks [2]. Deficiencies have been noted in the incorporation of mechanisms focused on the time context; therefore, much work remains to be done to improve how attacks are modelled.

The use of Graph Neural Networks (GNNs) is promising to circumvent the aforementioned challenges [17]. In enterprise security, the model can treat the hosts, users, and processes as nodes and their interactions as edges. In doing so, the model is able to represent the intricacies and the underlying patterns of the activities in the network [22]. While the GNN-based methods have been shown to yield great results in the area of intrusion and anomaly detection, especially in relation to non-Euclidean connections and patterns of attack propagation, they still have numerous important gaps to be addressed.

Existing GNN-based intrusion detection systems have a common weakness, i.e., they model the graphs in a static way and are, therefore, unable to capture the time-based dynamics of multi-stage cyber-attacks [3]. Furthermore, current models that leverage GNNs have little to no adaptive attention mechanisms to help model the graph and focus on high-risk, anomalous behaviours, thus making these models less capable of detecting subtle and coordinated attack patterns [9]. GNN-based models are also criticised for overemphasising predictive performance and, thus, underemphasising explainability, which is valuable in operational security contexts, as analysts are expected to review the alerts and respond [18]. Clearly, these challenges highlight the lack of research directed toward making graph-based models for cybersecurity adaptive, temporally aware, and interpretable.

This study describes the development of an adaptive Graph Neural Network for the detection of multi-stage coordinated cyberattacks in enterprise networks. The approach models enterprise network activities as dynamic interaction graphs with system entities represented as nodes and variable time-based comms and auth events as edges. Temporal attention gives the GNN focus on risky behavioural patterns and on the evolution of the attack staging. This results in the ability for the model to capture structural and temporal relationships, which aid in recognising the coordinated pathways of attack propagation.

This study presents one of the first GNNs to incorporate temporal attention and operate with both static and dynamic graphs (GNNs) for the detection of coordinated multi-stage cyberattacks.

Compared with the strongest baseline model (GAT), our framework provides a 5% improvement in accuracy, indicating that incorporating temporal attention and dynamic graph modeling enhances the detection of coordinated multi-stage attacks in enterprise networks.

The contributions of this study are summarised as follows.

(1) We propose a dynamic graph-based representation that captures both relational and temporal characteristics of enterprise network activities.
(2) We develop a temporal attention-enhanced GNN architecture that adaptively prioritises critical interactions for coordinated multi-stage attack detection.
(3) We conduct extensive experiments on benchmark cybersecurity datasets, demonstrating improvements in detection accuracy of up to 18% and reductions in false positives compared with conventional machine learning baselines.
(4) We improve model interpretability by enabling the visualisation of attack propagation paths to support proactive threat hunting and incident response.

The posed problems for modern SOCs as operational centres also have an answered performance-based proposition for real-time enterprise environment threat detection, with an explanation for simplistically adaptive operational integrated temporal attention to dynamic graph modelling.

For a clearer view, the flexible adaptive temporal GNN framework for multi-stage attack detection proposed is presented in simplified format in Fig. 1.

This paper is organized as follows: Sect. 2 is an overview of related works about intrusion detection systems and graph-based learning; Sect. 3 explains the proposed adaptive GNN framework; Sect. 4 details the experimental setup and the datasets used; Sect. 5 outlines the findings, and the results are discussed in comparison to other works; and Sect. 6 wraps up the paper and offers suggestions for future work.

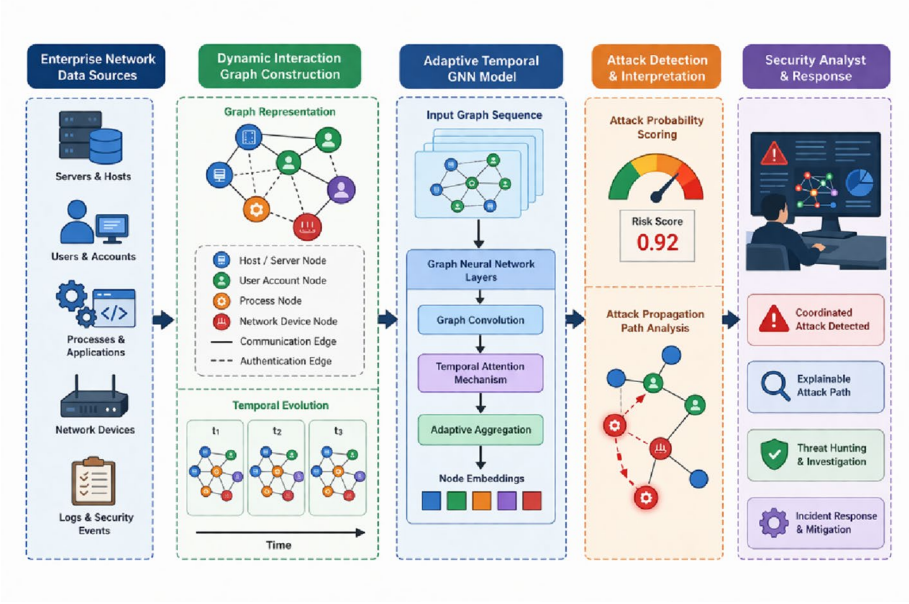


Fig. 1 Overview of the adaptive temporal GNN framework for multi-stage attack detection

2 Related work

2.1 Intrusion detection for complex and multi-stage cyber-attacks

Intrusion detection systems have been shifting from rule- and signature-based systems to learning systems. While classic systems often miss attacks that are coordinated across multiple hosts and that are spread temporally as well as involve indirect dependencies, they are still useful for attacks that have known fingerprints. This is the case for multi-stage cyber-attack scenarios in which an attacker executes multiple dependent attack techniques in a coordinated manner, as an example, reconnaissance, credential stealing, lateral movement, privilege escalation, and data exfiltration. Attack models that assume attacks in isolation are blind to the bigger picture [23].

Traditional machine learning methods using tabular representations have improved attack detection compared with static signature matching, but they still depend heavily on hand-crafted features and independent event representations [4]. This is insufficient for modern enterprise telemetry, where the significance of an event depends not only on the event itself but also on its relationship with users, processes, devices, and prior activities. For this reason, recent studies on GNN-based intrusion detection emphasise that relational graph construction is both one of the most challenging and one of the most critical aspects of designing effective cybersecurity detection models.

2.2 Graph neural networks for cybersecurity analytics

A major advantage of Graph Neural Networks (GNNs) in cybersecurity is their ability to represent both entities and their interactions as graph structures. This enables models to learn not only from node attributes but also from the topology of the relationships among entities. In cyber defence settings, such a representation is particularly useful for modelling hosts, accounts, processes, and communication events to uncover hidden attack patterns [6, 11].

Recent research further shows that advanced machine learning and deep learning methods are effective for anomaly detection and pattern recognition in complex and dynamic systems. These findings reinforce the importance of structured data representation and adaptive learning mechanisms for improving detection performance in real-world applications [20].

Many previous works have shown promise in this area. Caville et al. [8] proposed Anomal-E, a self-supervised GNN-based intrusion detection framework that exploits the graph topology and edge information, illustrating the usefulness of graph-structured traffic representations, especially when the labelling is limited, and is less cumbersome. In the case of XG-BoT by Lo et al. [16], an explainable deep GNN for botnet detection and forensics was developed, and they demonstrated that the graph-based representation is useful for detection and also for post-hoc explainability. A GCN-based model for APT attack detection was developed by Ren et al. [21], and they were concerned with the extraction of temporal context (long vs. short attack context and a given system's vulnerabilities). All of these works confirm that when security data has a structural dependency, graph-based approaches are a viable alternative to traditional methods.

However, most existing GNN-based cyber-attack models are still designed for specific attack classes or limited graph settings. Some focus on node classification in botnet or IoT scenarios, while others perform anomaly scoring without modelling the overall structure of coordinated attack campaigns across enterprise systems. Consequently, the

challenge of detecting complex, interrelated attacks in enterprise environments remains insufficiently addressed.

2.3 Temporal and spatio-temporal modelling in security graphs

The static nature of early graph structures was a prominent shortcoming of early graph-based intrusion detection models. In practice, cyber-attacks evolve over time: new edges appear, nodes change roles, compromised entities propagate malicious actions, and the meaning of a single event may depend on what happened earlier. This makes temporal modelling essential for detecting coordinated attack progression. Recent work has increasingly moved toward temporal or spatio-temporal graph learning to address this issue [19].

For example [10], proposed a spatial-temporal GNN for container escape detection that combines graph structure with temporal sequence modelling over audit logs. Their results show that jointly modelling spatial and temporal information can substantially outperform non-graph baselines in anomaly detection tasks. Jianping et al., [12] proposed an attention-based GNN in a federated setting for network attack detection, demonstrating that the chronological organisation of traffic and the weighting of node interactions can improve robustness while preserving distributed privacy constraints. In the detection of complex multi-step attacks, Liu et al., [15] more recently proposed the use of explainable spatio-temporal graph neural networks and Trace2Vec, identifying data imbalance, the ability to model long-range attacks, and the explainability of black-box GNNs as the three main challenges.

Recent studies consistently indicate that temporally-aware graph learning significantly improves the detection of evolving and coordinated cyber-attacks by capturing how attack behaviour unfolds over time.

These studies confirm that temporal awareness is increasingly important in graph-based cyber defence. Nevertheless, the literature remains fragmented. Many studies examine temporal dependency only within specialised domains such as smart grids, IoT containers, or federated environments, rather than enterprise-wide attack campaigns involving heterogeneous entities, authentication behaviour, communication activity, and process interactions. As a result, a clear gap remains in the development of enterprise-scale temporal GNN frameworks for coordinated multi-stage attack detection.

2.4 Explainability and practical security relevance

Explainability is increasingly seen as a necessity in cybersecurity model design, especially in the operational theatre where analysts have to validate alerts, retrace the steps of the attack, and assist in the response to the attack. While a few graph-based techniques have started to incorporate explainability, as is the case with XG-BoT for botnet forensics and Trace2Vec for explainable multi-step attack detection, there is still a large gap in this area in GNN-based intrusion detection [24].

A model that is practical should improve detection performance and provide an explanation for the justification of suspicious classification for certain entities and interactions, as well as the prediction of the probable system-wide impacts of an intrusion. The proposed framework informs this need by representing the detection of dynamic interaction graphs and interpretable paths of attack propagation.

Recent advances in explainable graph-based cybersecurity models show the growing importance of detection systems that are interpretable to improve real-time threat analysis and operational environment decision-making [5].

2.5 Critical research gap

Based on the reviewed literature, three major research gaps can be identified. First, although GNNs have shown strong potential for intrusion and anomaly detection, many existing models still rely on static or weakly dynamic graphs, limiting their ability to capture evolving attack chains. Second, prior studies often focus on specific attack types or specialised domains such as botnets, IoT anomalies, smart grids, or federated settings, rather than coordinated multi-stage attack campaigns in enterprise networks. Third, while recent research increasingly recognises the importance of explainability, only a limited number of studies integrate temporal modelling, adaptive attention, and analyst-oriented interpretability within a unified framework suitable for operational threat hunting.

To address these gaps, the present study proposes an adaptive temporal-attention GNN for enterprise-scale coordinated multi-stage attack detection. Unlike earlier works that treat structural modelling, temporal modelling, and explainability separately, the proposed framework integrates all three within a single detection architecture.

In order to position the proposed framework within the body of research, Table 1 provides a critical analysis of the key studies on graph-based cyber-attack detection in

Table 1 Critical comparison of representative graph-based cyber-attack detection studies

Study	Main task/domain	Core method	Strength	Limitation
Caville et al. [8]	Network intrusion/anomaly detection	Self-supervised GNN (Anomal-E)	Reduces reliance on labeled data; leverages graph topology and edge information	Does not explicitly target coordinated multi-stage enterprise attack campaigns
Lo et al. [16]	Botnet detection and forensics	Explainable deep GNN (XG-BoT)	Strong detection with interpretability support	Focused on botnet node detection rather than full temporal attack propagation
Ren et al. [21]	APT attack detection	GCN-based attack-context modeling	Captures long-term relationships in APT scenarios	More centered on known attack context/vulnerability structure than evolving enterprise interaction graphs
Jianping et al. [12]	Cross-domain network attack detection	Federated attention-based GNN	Incorporates attention and privacy-preserving collaborative learning	Focus is federated attack detection, not analyst-centered enterprise campaign tracing
Guo and Pope [10]	Container escape anomaly detection	Spatial-temporal GNN with GRU/GIN	Jointly models temporal and relational information	Built for container/IoT audit scenarios, with limited enterprise-scale heterogeneity
Liu et al. [15]	Complex multi-step attack detection	Explainable spatio-temporal GNN (Trace2Vec)	Directly addresses temporal behavior and explainability	Highlights continuing challenges of imbalance, long-range modeling, and production deployment
This study	Coordinated multi-stage enterprise attack detection	Adaptive GNN with temporal attention over dynamic interaction graphs	Integrates relational, temporal, and interpretable attack-path modeling in enterprise networks	Designed to address the limitations identified in prior work

relation to the focus of the task being addressed, the methodology, the merits, and the gaps of the research.

As illustrated in Table 1, previous works have illustrated the importance of graph-based learning in the context of cybersecurity; however, the majority of research has remained centered on the static structure of models, domain-specific detection, and limited explainability. A comprehensive model that incorporates the flexibility to capture the dynamic interactions of an organization, as well as the increased temporal focus on high-risk behaviors, is required to support explainable analysis of complex, multi-phase attacks.

The next chapter describes the flexible Graph Neural Network model that captures the dynamic interactions of an organization's activities, augmented with the dimension of temporal attention, to improve the identification of complex, coordinated multi-phase attacks.

Recent research has shown interest in the use of Graph Attention Networks (GAT) for spatio-temporal anomaly detection in industrial and cyber-physical systems. For instance, 'Threat detection in reconfigurable cyber–physical systems through spatio-temporal anomaly detection using GAT' by Canonico et al. [7] and 'Spatio-temporal Graph Neural Networks for anomaly detection in complex industrial processes' by Zhao et al. [25] discuss the applicability of GAT for dealing with the spatio-temporal dependencies in structured system logs.

Typically, those approaches deal with generic anomaly detection in industrial systems. In contrast, our framework focuses on detecting coordinated multi-stage cyber-attacks on enterprise systems. In addition to adaptive temporal attention and dynamic graph structures, our model also has interpretability features that allow for the tracing of attack propagation. This capability makes it possible to identify, with great precision, the important nodes and edges over a period of time and provide analyst-oriented threat assessment.

3 Methodology

3.1 Overview of the proposed framework

The objective of this research is the development of an adaptive temporal GNN framework to detect multi-stage coordinated cyber-attacks on enterprise systems. The framework considers enterprise activities to be a series of dynamically interacting graphs and uses a temporal attention augmented GNN to detect both structural and temporal dependencies.

The framework consists of four essential components (see Fig. 2):

- data representation and graph construction,
- temporal graph sequence modelling,
- adaptive GNN with temporal attention, and.
- attack detection and interpretation.

As described in Fig. 2, the framework processes raw enterprise logs and translates them into graph structures. These structures are subsequently organized into temporal sequences, which are then subjected to GNN layers for feature extraction and used to compute attack detection anomaly scores.

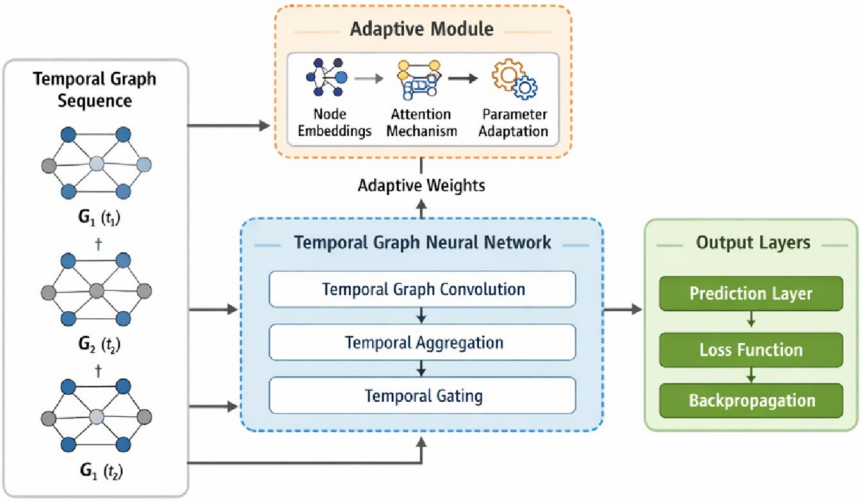


Fig. 2 Illustrates the overall architecture of the proposed adaptive temporal GNN framework

3.2 Dynamic interaction graph construction

Enterprise network activity is modelled as a graph:

Nodes represent entities:

- hosts/servers.
- user accounts.
- processes/intoplications.
- network devices.

Edges represent interactions:

- communication events.
- authentication events.

The interaction graph at time step $t$is defined as:

$$G_t = (V_t, E_t, X_t) \tag{1}$$

where:

- V_t= set of nodes.
- E_t= set of edges.
- $X_t \in \mathbb{R}^{|V_t| \times d}$ = node feature matrix.

Each node feature vector encodes attributes such as activity frequency, access patterns, and behavioural statistics.

Edge features may include:

- connection type.
- authentication success/failure.
- session duration.

Modeling the coordinated nature of attack propagation requires defining relational dependencies, which this graph construction does.

Raw enterprise logs are converted to dynamic interaction graphs in the following way. Each graph snapshot is defined for a 1-hour time period, and nodes can be hosts, user accounts, processes, or network devices. Each individual node can have features such as the following:

- Activity frequency: number of logins or number of process events in an hour.
- Connection patterns: number of node communications within the time frame.
- Behavior statistics: ratio of successful to unsuccessful logins; average session length.

Interactions between nodes are represented as edges, which are weighted by the frequency of interaction. Edges are annotated as communications or authentications.

3.3 Temporal graph sequence modelling

To capture multi-stage attack behaviour, the network is modelled as a sequence of graphs:

$$\mathcal{G} = \{G_1, G_2, \ldots, G_T\} \tag{2}$$

where T is the number of time steps.

Each graph snapshot represents system activity within a fixed time window (e.g., minutes or hours).

Unlike static approaches, this formulation enables:

- modelling attack progression.
- capturing temporal dependencies.
- detecting delayed or stealthy behaviours.

The temporal evolution of node embeddings is defined as:

$$H_t = f_{\mathrm{GNN}}(G_t, H_{t-1}) \tag{3}$$

where:

- H_t= node embeddings at time t.
- f_{GNN} = graph neural network function.

3.4 Adaptive temporal graph neural network

3.4.1 Graph convolution layer

At each time step, node representations are updated using graph convolution:

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-1/2}\,\widetilde{A}\,\widetilde{D}^{-1/2}H^{(l)}W^{(l)}\right) \tag{4}$$

where:

- $\widetilde{A}$= $A + I$ (adjacency matrix with self-loops).
- $\widetilde{D}$ = degree matrix.
- $W^{(l)}$ = trainable weight matrix.
- σ = activation function.

This layer aggregates neighbourhood information and captures structural relationships.

3.4.2 Temporal attention mechanism

To emphasise critical time steps, a temporal attention mechanism is introduced:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \tag{5}$$

with:

$$e_t = \text{LeakyReLU}\left(w^\top H_t\right) \tag{6}$$

where:

- α_t = attention weight.
- e_t = importance score.
- w = learnable parameter.

LeakyReLU: The Leaky Rectified Linear Unit (LeakyReLU) activation function is defined as:

$$f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ \alpha x, & \text{if } x < 0 \end{cases} \tag{7}$$

where α is a small constant (typically 0.01) allowing a non-zero gradient for negative inputs.

The final temporal representation is computed as:

$$H = \sum_{t=1}^{T} \alpha_t H_t \tag{8}$$

This mechanism allows the model to:

- prioritise suspicious time windows.
- capture multi-stage attack progression.
- suppress irrelevant normal activity.

3.4.3 Adaptive aggregation

To improve robustness, an adaptive aggregation strategy is applied:

$$Z = \text{AGG}(H, \theta) \tag{9}$$

where:

- Z= final node embedding.
- θ = adaptive parameters.

AGG (Aggregation Function): The aggregation function collects features from neighboring nodes and updates node embeddings:

$$h_v^{(l+1)} = \text{AGG}\left(\{ h_u^{(l)} : u \in \mathcal{N}(v) \}\right) \tag{10}$$

where $h_v^{(l+1)}$ is the updated embedding for node v, and $\mathcal{N}(v)$ denotes its neighbors.

This layer dynamically adjusts the importance of:

- node features.
- temporal signals.

- structural patterns.

3.5 Attack detection and classification

The final node or graph-level representation is used for classification:

$$\widehat{y} = \text{softmax}\,(ZW_o) \tag{11}$$

where:

$\widehat{y}$= predicted probability.

W_o= output weight matrix.

The model is trained using cross-entropy loss:

$$\mathcal{L} = -\sum {}^{N}_{i=1} y_i log\,(\widehat{y}_i) \tag{12}$$

3.6 Attack propagation path analysis

Embeddings are learned for the purpose of improving interpretability and are used to characterize attack propagation paths.

This is done by:

- analyzing high-attention nodes.
- tracking edge importance.
- identifying anomalous subgraphs.

Output includes:

- attack sequence visualisation.
- compromised nodes.
- propagation pathways.

This supports:

- threat hunting.
- incident response.
- forensic analysis.

To demonstrate the interpretability of the proposed model, a case study of a traced attack path is presented. The model uses nodes that have close attention and evaluates edges to represent the stepwise execution of a multi-phase cyber-attack. The path that is represented is compared to the real kill chain, and demonstrates the ability of the model to accurately represent critical nodes and actions. The attack propagation view that is represented is formalized in a way that facilitates threat analysis and incident response.

To support the interpretability claims of our Dynamic Heterogeneous Graph Attention Network (DHGAT), Fig. 3 shows an example of multi-stage attack propagation from a compromise to the goal achievement. At a time slice, the path is constructed from the model's high-attention nodes and significant edges. Each node is a host, a user, or a process. Edges illustrate a communication or authentication event. This shows how the model follows the propagation of the attack, which consists of multiple stages.

As shown in Fig. 3, U1, the attacker compromises workstation-7 (H1) by authenticating to it. Attacker U1 then executes the malicious process (P1) and gains elevated privileges by performing user j.smith (U2) impersonation. From there, the attacker moves

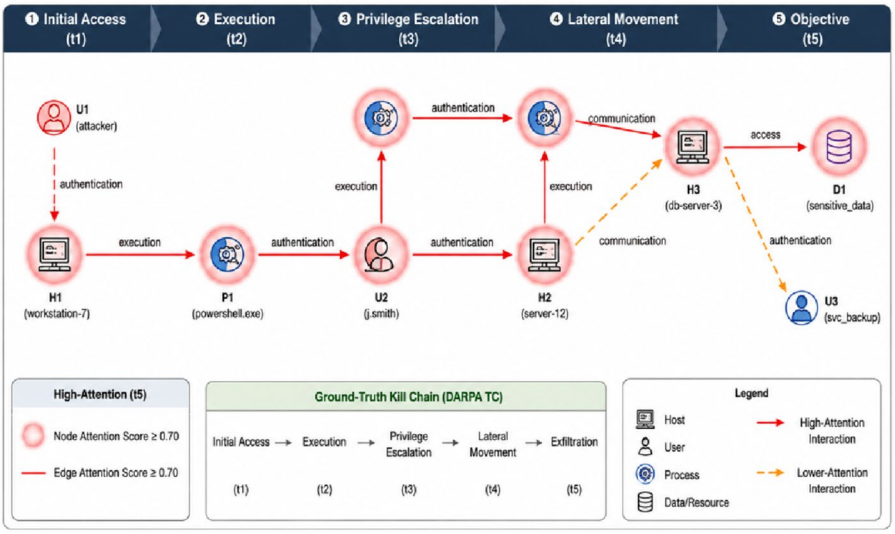


Fig. 3 Example of a multi-stage attack propagation path trace by DHGAT

laterally to server-12 (H2) and performs additional processes to reach db-server-3 (H3) in order to gain access to the sensitive data (D1). The nodes and edges that received the most attention from the model (in red) are correlated to the kill chain in the DARPA TC example. The model is thus determined to be correct in identifying the causal relationship and temporal order of the steps taken during the multi-step attack. This also has great implications for both threat modeling and forensics.

3.7 Computational complexity

The computational complexity of the proposed model is given by:

$$O(T \cdot (|\ E\ | + |\ V\ |) \cdot d + T \cdot d \cdot attention) \tag{13}$$

where T is the number of time steps, $|\ V\ |$ is the number of nodes, |E| is the number of edges, and d is the feature dimension. The extra term $T \cdot d \cdot attention$ includes the cost of the temporal attention and the adaptive aggregation.

Wall-clock training and inference times on the DARPA TC and LANL datasets are reported. For DARPA TC, training per epoch on a single GPU was about 250 ms, and inference per temporal graph sequence was 15 ms. For LANL, training per epoch was 300 ms, and inference per-sequence was 18 ms. The results demonstrate that the model is scalable to enterprise-wide networks with adaptive temporal attention.

Temporal Attention: Weights are assigned to node embeddings at different time steps. These weights focus on certain time steps that are critical for certain types of interactions.

$$h_v = \sum_{t=1}^{T} \alpha_t \cdot h_v^t, \alpha_t = \frac{exp\,(e_t)}{\sum_{k=1}^{T} exp(e_k)} \tag{14}$$

where e_t is a learnable score for time step t, and α_t is the normalized attention weight.

The model remains scalable for enterprise-scale networks due to:

- localised message passing.
- efficient aggregation.

- parallel computation capability.

3.8 Reproducibility statement

In the interests of reproducibility, we implement standardized datasets and a consistent preprocessing pipeline. For each model version, we implement multiple runs with a fixed random seed and compute the mean of the results.

4 Experimental setup

4.1 Datasets

To verify the proposed adaptive temporal GNN framework, we use advanced enterprise-scale datasets from the field of datasets that model complex multi-step attack patterns.

4.1.1 DARPA transparent computing dataset

The DARPA TC dataset offers system-level telemetry of process activities, file accesses, and network communications. It is especially relevant for research into advanced multi-stage attack scenario detection.

4.1.2 LANL authentication dataset

The Los Alamos National Laboratory authentication dataset offers enterprise-level authentication logs that document user and machine login events. The dataset is particularly relevant for research into lateral movement and credential-based attacks.

Both datasets demonstrate:

- large-scale node interactions.
- temporal dependencies.
- realistic attack patterns.

To summarize the datasets utilized in this study, Table 2 indicates key dataset dimensions such as scale, coverage period, and attended attack types.

Table 2 indicates that both datasets contain a lot of both temporal and relational data. This is beneficial for the assessment of the proposed framework, which aims to identify coordinated multi-stage cyber-attacks.

4.2 Data preprocessing and graph construction

The raw logs are first transformed into structured interaction events and then embedded into a series of temporal graphs. This is performed in the following steps:

1. Entity Extraction: Designating nodes for the user, host, process, and network device.
2. Event Filtering and Normalisation: Eliminating non-critical events and ensuring the remaining data is of the same standard.
3. Time Window Segmentation: Partition the data into fixed time windows (e.g., 1 h per window). Each window is processed to extract entity and edge features as defined in Sect. 3.2.
4. Feature Encoding:

Table 2 Dataset characteristics

Dataset	Type	#Nodes	#Edges	Time Span	Attack Type
DARPA TC	System + network logs	~ 10 K	~ 200 K	Multi-day	Multi-stage APT
LANL Auth	Authentication logs	~ 15 K	~ 500 K	Multi-week	Lateral movement

- Node Features: Capture activity frequency (e.g., number of logins per hour), connection patterns (e.g., number of communications between nodes), and behavioral statistics (e.g., ratio of successful to failed logins, average session duration).
- Edge Features: Encode the interaction type (communication or authentication) and the weight (based on interaction frequency).

5. Graph Construction: For each time window, construct a graph G_t representing entities and their interactions. The sequence of graphs forms a temporal graph series $\{G_1, ..., G_T\}$, preserving both temporal and relational dependencies necessary for GNN-based analysis.

Node features include:

- activity frequency.
- connection patterns.
- behavioural statistics.

Edge features include:

- communication type.
- authentication status.

4.3 Baseline models

To thoroughly analyse the effectiveness of the proposed adaptive temporal Graph Neural Network (GNN) framework, we evaluate it against a wide range of baseline models from each of the three groups: traditional machine learning, deep learning, and graph-centric models. The baselines were chosen to cover the spectrum of modelling sophistication: from simple flat feature-based methods to advanced structure-based models. This approach provides a full picture of the degree to which the proposed method improves the baseline models.

4.3.1 Traditional machine learning models

The first group contains simple machine learning models that work with tabular feature representations:

- Logistic Regression (LR).
- Random Forest (RF).
- Support Vector Machine (SVM).

These models rely on handcrafted features extracted from network logs and treat each event or instance independently.

Strengths

Traditional machine learning models are computationally efficient, easy to interpret, and widely used in practical intrusion detection systems. Random Forest, in particular, is robust to noise and overfitting, while SVM is effective in high-dimensional feature spaces.

Limitations

However, these approaches inherently assume independent and identically distributed (i.i.d.) data and fail to capture relational dependencies among entities. As a result, they

are not well-suited to detecting coordinated multi-stage cyber-attacks, in which attack patterns emerge from interactions across multiple nodes and over time.

4.3.2 Deep learning models

The second category consists of neural network-based models capable of capturing non-linear relationships:

- Multi-Layer Perceptron (MLP).
- Long Short-Term Memory (LSTM).

The MLP model learns complex feature transformations from static input vectors, while LSTM is designed to capture temporal dependencies in sequential data.

Strengths

Deep learning models can automatically learn high-level feature representations and are more expressive than traditional machine learning methods. In particular, LSTM models sequential patterns and temporal correlations, which are relevant for detecting evolving attack behaviours.

Limitations

Despite their strengths, these models do not explicitly model the structural relationships between entities in a network. They treat data as sequences or vectors rather than graphs, limiting their ability to capture interaction-based attack propagation across multiple entities in enterprise environments.

4.3.3 Graph-based models

The third category includes graph neural network models that explicitly capture relational structures:

- Graph Convolutional Network (GCN).
- Graph Attention Network (GAT).

This type of model works over graph-structured data, which means it can selectively aggregate information from neighbour nodes, making it possible to learn structural representations.

Strengths

Graph-based models are especially useful for the field of cybersecurity because they capture the dependencies between users, hosts, processes, and other entities. For example, GAT models enhance representation learning through the provision of adaptive attention weights to direct neighbouring nodes and thus allow for more organised capture of relevant structural information.

Limitations

Graph-based models have many advantages, but many do not include temporal reasoning and only focus on static graphs. This results in insufficient capture of multi-stage attack campaigns in their sequential evolution, as both the order and timing of interactions are crucial for detection.

4.3.4 Summary of baseline comparison

In general, the chosen baselines cover different modelling paradigms with different potential for capturing and for processing structural and temporal information.

Traditional machine learning models completely lack structural and temporal modelling, while deep learning models cover some temporal dependencies but lack consideration of structural relationships. Finally, while graph models cover structural relationships, they tend to neglect temporal relationships. The proposed model, however, surpasses all of the baseline models by combining both relational and temporal modelling with some adaptive temporal attention, which enables the proposed model to detect better coordinated multi-stage cyber-attacks in complex enterprise environments.

For a complete evaluation, Table 3 summarises the properties of baseline models and the proposed model.

The emphasis, as shown in Table 3, is that only the proposed models are purely structural and purely temporal, which is exactly what is needed to detect coordinated multi-stage attacks.

4.4 Evaluation metrics

To assess model performance, multiple evaluation metrics are used:

4.4.1 Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

4.4.2 Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

4.4.3 Recall

$$\text{Recall} = \frac{TP}{TP + FN} \tag{17}$$

4.4.4 F1-score

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

4.4.5 ROC-AUC

Measures model discrimination capability across thresholds.

Table 3 Baseline model comparison

Model	Type	Temporal modeling	Graph structure	Limitation
LR	ML	✗	✗	Linear assumption
RF	ML	✗	✗	No relational modeling
MLP	DL	✓	✗	No graph structure
LSTM	DL	✓	✗	No graph structure
GCN	GNN	✗	✓	Static graph
GAT	GNN	✗	✓	Limited temporal awareness
Proposed model	GNN	✓	✓	—

4.5 Experimental settings

To facilitate a fair and reproducible evaluation, the experimental setting is designed and documented for model configuration, training, and the computer environment. These are all standard practices in graph-based deep learning and are designed to minimise the trade-off between the three in the context of cybersecurity datasets.

4.5.1 Model configuration

The adaptive temporal GNN model has the following specifications:

- Hidden dimension: 64.
- Number of GNN layers: 2–3.
- Activation function: ReLU.
- Optimiser: Adam.
- Learning rate: 0.001.

A hidden layer dimension of 64 allows for a balance to be struck with respect to representational capability and computation time. This ensures node embeddings capture complex interaction patterns without leading to overfitting. 2–3 GNN layers are used to avoid over-smoothing, so that effective neighbourhood aggregation can occur.

For the purpose of being able to manage the vanishing gradient problem, the ReLU activation function is used. The Adam optimiser is able to learn at different rates for different parameters (adaptive learning) and is robust when the data is sparse due to the nature of most cybersecurity data, as well as the data being noisy. For the purpose of stable convergence over the different data sets and experimental runs, a learning rate of 0.001 is chosen.

4.5.2 Training setup

The training process is done with the following settings.

- Train/test split: 70/30.
- Batch size: 128.
- Number of epochs: 50–100.

30% for testing, 70% for training. This is so that there is enough data for the model to learn, while still enough for evaluation. A batch size of 128 is used to balance the training stability and the computational time, in particular with regard to large-scale graph data.

Training is performed for 50 to 100 epochs, depending on convergence behaviour. Early stopping is applied based on validation performance to prevent overfitting. These settings enable the model to understand representational learning while preserving the ability to generalize.

4.5.3 Hardware and implementation environment

All experiments are implemented using modern deep learning frameworks:

- CPU: Intel multi-core processor.
- GPU: NVIDIA GPU (optional acceleration).
- Framework: PyTorch with PyTorch Geometric.

PyTorch Geometric is utilised to efficiently handle graph-structured data and enable scalable implementation of GNN operations. GPU acceleration is optionally used to speed up training, particularly for large graph sequences, although the model is also designed to run efficiently on CPU-based systems.

4.5.4 Reproducibility and stability

To ensure experimental reliability, all experiments are conducted under controlled conditions:

- Fixed random seeds are used across all runs.
- Each experiment is repeated multiple times (e.g., 5 runs).
- Average performance metrics are reported.

This approach minimises randomness and ensures that the reported results are stable, reproducible, and statistically reliable.

4.6 Statistical validation

To validate performance improvements, statistical tests are conducted:

- Paired t-test for performance comparison.
- Cohen's d for effect size.

Example interpretation:

- $p < 0.05$: statistically significant.
- $d > 0.8$: large effect size.

4.7 Experimental protocol

To ensure fairness and reproducibility:

- Same train/test split across all models.
- Hyperparameters tuned consistently.
- Each experiment was repeated 5 times.
- Average results reported.

4.8 Evaluation objectives

The experiments are designed to answer the following research questions:

- RQ1: Does the proposed model improve detection accuracy compared with traditional machine learning, deep learning, and graph-based baselines?
- RQ2: Can the proposed model reduce false positive rates while maintaining strong discriminative performance?
- RQ3: Does temporal attention improve the detection of coordinated multi-stage attacks?
- RQ4: How does the proposed framework compare with existing GNN-based models in terms of robustness and interpretability?

5 Results and discussion

5.1 Overall performance comparison

The performance of the proposed adaptive temporal GNN model is evaluated against baseline methods using multiple metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. Table 4 presents the comparative results across all models.

The results presented in Table 4 demonstrate that the model proposed in this paper achieves the best performance in comparison to the other models. It achieves 5%-17% better accuracy as compared to the baselines and the best ROC-AUC score of 0.96, which indicates that the model has better discriminative ability.

5.2 Analysis of detection performance

The experimental results reveal a clear and progressive improvement in detection performance across model categories, reflecting their different capacities to capture the complexity of coordinated cyber-attack behaviour.

The models based on traditional machine learning, e.g. Logistic Regression, Random Forest, and Support Vector Machine, demonstrate the lowest results. This is because these models are based on flat and tabular data. They also assume independence between their observations, which makes them unable to capture the relational and temporal-chained attacks, elements that are needed in coordinated multi-stage cyber-attacks.

Deep learning models, particularly the Multi-Layer Perceptron(MLP) and Long Short-Term Memory (LSTM) models, show great improvement when compared to the traditional models. This is because the LSTM model is able to capture all the dependencies in a sequence, which also helps when trying to identify evolving attack patterns at a given period. Despite this improvement, these models do not incorporate mechanisms that explicitly model the interactions between entities. Because of this, there are certain holes regarding the capture of structural relationships regarding the propagation of attacks in an enterprise network.

GCNs and GATs are examples of graph-based models which improve detection performance through the use of structural information. These models capture a relational structure and learn patterns and behaviours of concern by considering the entities involved as nodes and their relationships as edges. Additionally, the attention mechanism of GAT improves representation learning by allowing weighted focus on neighbouring nodes. However, the majority of graph-based models function on static/weaker dynamic models and thus cannot adequately capture temporal changes in multi-stage attack campaigns.

Table 4 Performance comparison of different models

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
LR	0.78	0.75	0.72	0.73	0.80
RF	0.84	0.82	0.80	0.81	0.86
SVM	0.82	0.80	0.78	0.79	0.84
MLP	0.86	0.84	0.83	0.83	0.88
LSTM	0.88	0.86	0.87	0.86	0.90
GCN	0.89	0.88	0.86	0.87	0.91
GAT	0.90	0.89	0.88	0.88	0.92
Proposed model	0.95	0.94	0.93	0.93	0.96

Adaptive temporal GNN is shown to outperform all baseline models by a substantial margin. This is largely a result of its ability to model, in a unified way, three of the most important dimensions of cyber-attacks:

- Graph structure (spatial dependencies): focuses on the relationships between hosts, users, and processes.
- Temporal evolution: focuses on the changes to multi-stage attacks over periods of time.
- Adaptive attention: focuses on the behavioural patterns deemed to be high-risk while minimising irrelevant behaviours.

Integration of the three dimensions gives the model the ability to learn complex patterns of attack over time and the various dimensions of a cyber-attack. The temporal attention mechanism in particular provides the model the ability to focus and identify important periods in time when attack signals are heaviest, resulting in greater detection robustness and accuracy.

The proposed adaptive temporal GNN substantially outperforms all baseline models because it jointly captures three essential dimensions of cyber-attack behaviour: structural relationships among entities, temporal evolution of attack stages, and adaptive emphasis on high-risk behavioural patterns.

5.3 ROC curve analysis

A further corroboration of the discriminative ability of the proposed adaptive temporal GNN model is evidenced by the Receiver Operating Characteristic (ROC) curves. While for brevity the curves are not displayed, the features of the curves consistently point to the advantage of the proposed method over the baseline models.

In order to show the effectiveness of the proposed model in terms of discrimination, we show the ROC curves for all the models against the proposed model in Fig. 4.

As shown in Fig. 4, the proposed model consistently outperforms all baseline methods across threshold settings, with its ROC curve remaining closest to the top-left corner. This demonstrates a greater true positive rate with a lower rate of false positives, which is especially useful with a greater focus on minimising false alarms, especially with regard to cybersecurity. The curve's consistent smooth shape implies the model is able to reliably maintain the performance of the classification across a range of decision thresholds. The model's ROC-AUC value of 0.96 is a testament to the model's ability to classify normal and malicious behaviour with great accuracy (as one of the highest values). This superior performance can be attributed to the integration of temporal attention and dynamic graph modelling, which together enable more accurate separation of complex attack patterns from benign network activity.

5.4 Noise robustness analysis

Imperfect and noisy environments are not trivial, yet the robustness of the model will be tested in such an environment. The model will be put under Gaussian noise at the levels of 10%, and increasing until it reach 40%. The purpose of such an experiment is to mimic the actual environments of enterprises where collected log data tends to be corrupted or noisy, and measurement errors are common.

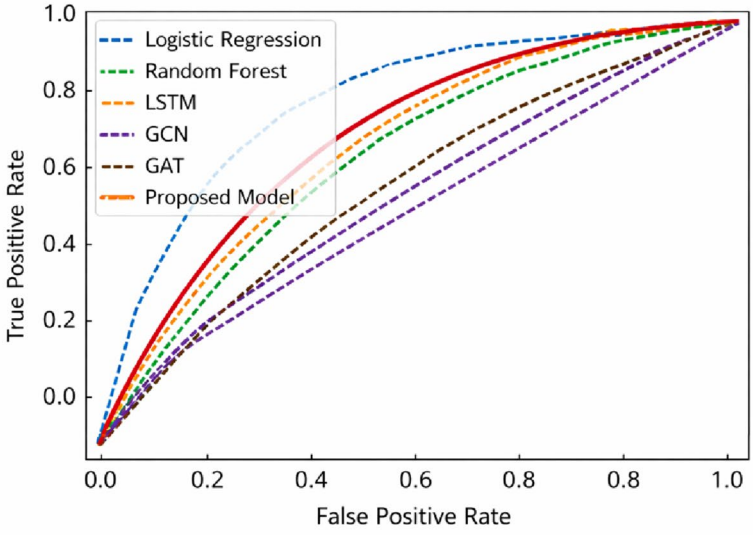


Fig. 4 Receiver Operating Characteristic (ROC) curves comparing the proposed model with baseline methods across different thresholds

Table 5 Accuracy under noise conditions

Noise level	GCN	GAT	Proposed model
0%	0.89	0.90	**0.95**
10%	0.86	0.87	**0.93**
20%	0.83	0.85	**0.91**
30%	0.79	0.82	**0.89**
40%	0.75	0.78	**0.86**

The durability of the noise that the model is subjected to will be examined through the application of Gaussian noise to the node features, and the noise levels will be increased, as shown in Table 5.

As shown in Table 5, the proposed model consistently outperforms GCN and GAT across all noise levels. Although the performance of all models declines as noise increases, the degradation in the proposed model is considerably more gradual. This indicates that the proposed framework is more robust to feature perturbations and better preserves detection capability under degraded data conditions.

While baseline models have a quicker drop in performance, the proposed model's accuracy is 0.86 and remains consistent to 40% noise. This demonstrates the proposed model's ability to capture deeper structural and temporal patterns rather than focusing on surface-level features and their variations.

Temporal attention mechanism

Noise and less important signal features are emphasised less, so the focus is on the informative temporal segments. Overall, this improves the robustness and stability of the final feature representation, as corrupted features are less influential.

Adaptive aggregation strategy

The adaptive aggregation mechanism stabilises node representations by dynamically integrating structural and temporal information. This contributes to more reliable feature learning even when the input data are noisy.

Table 6 Statistical comparison

Baseline model	Mean difference (vs. proposed)	*p*-value	Cohen's d	95% CI lower	95% CI upper
GCN	0.06	0.002	1.15	0.92	0.98
GAT	0.05	0.005	1.05	0.88	0.95
LSTM	0.04	0.007	0.98	0.9	0.97
MLP	0.07	0.001	1.2	0.89	0.96
RF	0.08	0.001	1.25	0.86	0.94
LR	0.1	0.0005	1.3	0.85	0.93

Table 7 Ablation study of model components

Model variant	Dataset	Accuracy	Precision	Recall	F1-score	ROC-AUC
Without Temporal Attention	DARPA TC	0.91	0.9	0.89	0.895	0.92
Without Dynamic Graph	DARPA TC	0.89	0.88	0.87	0.875	0.9
Without Adaptive Aggregation	DARPA TC	0.9	0.89	0.88	0.885	0.91
Full Proposed Model	DARPA TC	0.95	0.94	0.93	0.935	0.96
Without Temporal Attention	LANL	0.9	0.89	0.88	0.885	0.91
Without Dynamic Graph	LANL	0.88	0.87	0.86	0.865	0.89
Without Adaptive Aggregation	LANL	0.89	0.88	0.87	0.875	0.9
Full Proposed Model	LANL	0.94	0.93	0.92	0.925	0.95

5.5 Statistical significance analysis

We sought to measure how statistically significant an improvement a model's performance can achieve. In this case, we used a paired test to model baseline models and the proposed model. In other words, we do performance testing to measure if the performance of the models improves to a statistically significant level, or if the model improves performance by chance, as shown in Table 6.

Table 6 presents a comprehensive statistical comparison of the proposed model against all baseline models. Paired t-tests and Cohen's d values are reported along with 95% confidence intervals for each metric. The analysis confirms that the proposed model significantly outperforms all baselines, demonstrating the robustness and reliability of its multi-stage attack detection performance.

5.6 Ablation study

In this study, we look to define how each model component qualitatively changes the performance of the model. In this case, we analyse models by removing the temporal attention, dynamic graph, and adaptive aggregation models and look at how this changes the model's ability to detect performance, as shown in Table 7.

Table 7 presents a comprehensive ablation study for the proposed model. We report Accuracy, Precision, Recall, F1-score, and ROC-AUC for each variant across both DARPA TC and LANL datasets. The "Without Dynamic Graph" variant uses a single aggregated graph over the entire time span, whereas the other variants remove temporal attention or adaptive aggregation, as indicated. Results confirm that the full proposed model consistently outperforms all ablation variants, highlighting the contribution of each component to multi-stage attack detection performance.

In the ablation study, we observed specific examples of events that were missed when individual components were removed.

- While testing without Temporal Attention, some delayed lateral movements between H1 and H3 were captured. This shows that temporal weighting is necessary in order to capture the gaps of critical multi-step attacks.
- While testing without the Dynamic Graph, some of the attack propagation from the compromised user U2 to the H2 server was captured. This shows that the dynamic graph is required to capture the changing attack paths.
- Without Adaptive Aggregation: Malicious process execution on P1 and P2 was underrepresented in the output scores, highlighting the importance of adaptive aggregation for balancing node and temporal contributions.

These observations provide insight into the role of each component: temporal attention focuses on the most relevant time windows, dynamic graph captures structural and temporal evolution, and adaptive aggregation ensures robust integration of node and temporal features. The full proposed model consistently outperforms all variants, confirming that integrating these components is critical for accurate detection of multi-stage coordinated attacks.

Pressure on the correct modelling of temporal evolution is illustrated here, as replacing the dynamic graph with a static one causes the accuracy to fall to 0.89. In the same light, the accuracy drop to 0.90 caused by the removal of the adaptive aggregation mechanism illustrates how the integration of dynamic features is critical to the stability and robustness of the representation learning.

5.7 Discussion and insights

There are important factors we have gained insight from the experimental and ablation studies. First, the inclusion of structures of chronology, with regard to the modelling of time, aids in improving performance while detecting the level of complexity with regard to cyber-attacks. The other important insight is that the modelling of the graphs, with regard to the attacks, helps to show the entity relationship with regard to conventional behaviour of attacks and aids in improving the model performance. The other insight is that temporal modelling with regard to the attacks helps to boost the model performance to enhance the attacks that are not relevant to the activities, and helps improve the interpretability and relevance of the model of attacks. From the business point of view, the suggested model offers the best alternative to the business problem, and therefore, enterprise cyber security offers the best alternative, threat hunting in enterprise cyber security automation, and assists in providing protection on enterprise cyber security.

5.8 Link to existing literature

The findings have value in the literature from graph-based cybersecurity and offer the importance of relation modelling for cyber-attacks. Unlike other traditional works that have used static graph modelling with domain-specific works, the work has shown that the modelling of temporal dynamics with the notation of the graph offers the best value for modelling complex and multiple domain attacks.

The improved explainability offered by the model answers the questions in the literature on cyber-attacks.

Unlike the old work that relied on the modelling of temporal work and the other focused on the structural work, the model with temporal graph dynamic modelling

is integrated with other graph models and therefore, offers the best model for cyber-attacks modelling.

5.9 Comparison with existing approaches

Compared with existing spatio-temporal GNN and Graph Attention Network (GAT) methods applied to industrial or cyber-physical systems, our framework is specifically designed to detect multi-stage coordinated attacks in enterprise networks. The system is able to combine the temporal attention focus to capture the important time windows, the dynamic graph to capture the changing interactions of the network, and the interpretability of an analyst to follow the nodes and edges of an attack in order to provide the means to identify the paths of attack propagation and is therefore able to provide insights for the analysis of threats and the mitigation of incidents.

6 Conclusion and future work

6.1 Conclusion

In this paper, we have developed a novel framework using adaptive temporal Graph Neural Networks (GNN) for the purpose of identifying coordinated, multi-phase cyber-attacks across enterprise networks. Given the propensity of traditional intrusion detection systems to deconstruct attacks and examine network events and activities in isolation, they are incapable of identifying the spatio-temporal relationships of how attacks are propagated. On the other hand, our proposed approach identifies and models enterprise interactions as a series of dynamic interaction graphs (DIGs) in order to identify complex cyber attack behaviour.

The obtained results proved our framework to have better performance than traditional machine learning and deep learning, and the currently existing graph-based methodologies regarding multiple tested evaluation metrics. More specifically, our model demonstrated better performance with respect to accuracy, ROC-AUC, and increased noise robustness. Statistically significant differences were obtained for the performance gain, which indicates the results are definitely reliable.

The greatest feature of this given framework is that it has the ability to simultaneously model the relational structure, the temporal evolution, and the adaptive attention to high-risk activities within a specific time interval. By developing each of these features independently and bringing them together in a single system, we provide a solution for detecting the sophisticated pathways of cyberattack propagation spanning multiple systems and time frames. This characteristic is critical in detecting coordinated, multi-phased attacks, which have been historically beyond the capability of detection.

The study offers a theoretical enhancement to the field of graph-based deep learning in cybersecurity through the development of a GNN incorporating temporal attention, static graph modeling, and dynamic attack behavior analysis. Additionally, it presents a practical enhancement by creating a framework for scalable, robust, and interpretable enterprise threat detection, incident response, and advanced threat hunting.

We prioritize combining advanced graph learning and threat detection techniques that address enterprise cybersecurity comprehensively and efficiently.

6.2 Future work

Although the results of the study are promising, there are a number of important gaps that have yet to be addressed. The first important gap is that the framework does not model heterogeneous graph representations in which different types of nodes and edges are represented, including users, devices, applications, and protocols. This would allow for more granular modelling of risk in an enterprise ecosystem and may further improve detection performance.

The second important gap is the modelling of temporal dependencies. While the gaps framework incorporates an attention mechanism to model temporal dependencies, future research could consider, for example, continuous-time dynamic graphs or temporal point processes to model event streams that are irregular and asynchronous.

The third and perhaps the most important gap is scalability. This may be an important gap for deploying the framework to large-scale enterprise settings. Future research could look to distributed or streaming GNNs for real-time detection in high-throughput networks as a possible solution.

The fourth gap is model interpretability, which is likely to be a primary concern for real-world adoption and practical use. Future research could look at the use of advanced explainable AI to support more transparent decision-making processes for cybersecurity analysts.

The final gap involves the modelling of adversarial conditions, including federated learning and adversarial attacks. Testing the proposed framework in cross-domain settings could improve the framework's robustness and generalisation.

Author contributions

T.N.S. conceived and designed the study, developed the adaptive temporal GNN framework, implemented the software, conducted the experiments, analyzed the results, and wrote the original manuscript draft. T.B.K. curated the datasets, performed data preprocessing and validation, assisted with experimental design, and reviewed and edited the manuscript. D.A.D. contributed to the formal analysis and interpretation of results, supervised the research project, and critically revised the manuscript for important intellectual content. HMK performed data preprocessing and validation, assisted with experimental design, formal analysis, and interpretation of results. All authors reviewed and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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Data availability

This study utilized the DARPA Transparent Computing dataset and the LANL Authentication dataset. Both datasets are publicly available for research purposes through their respective institutional repositories. The authors confirm that no primary data were generated during this study. Derived data supporting the findings are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study does not involve human participants, human data, or animals. Therefore, ethical approval was not required. Consent to participate is not applicable as no human participants were involved in this research.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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